ENTERPRENEX SOLUTIONS PRIVATE LIMITED

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Enterprise Workflow Management System (EWMS) • Release 2.0

Document Overview: This document provides the complete, authoritative Software Requirements Specification (SRS) for the **Enterprenex EWMS (Enterprise Workflow Management System)**. Formatted in accordance with the **IEEE Std 830-1998** and **ISO/IEC/IEEE 29148:2018** standards, this document specifies all functional requirements, non-functional constraints, 5-tier Role-Based Access Control (RBAC) governance, data schemas, API specifications, and operational workflows for the 14 institutional modules governing workforce operations, attendance, leaves, project governance, invoicing, compliance, and zero-trust IT administration.

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System / Product:	Enterprenex EWMS Suite	**Version / Status:**	v2.0 (Production Verified)
Author / Architecture:	Lead Systems Architect & Core Dev	**Approved By:**	Managing Director / CEO
Target Stack:	React 19 + TypeScript + Node.js + Prisma	**Classification:**	Institutional Confidential

Document Revision History

Version	Date	Author	Summary of Changes
v1.0	2026-10-05	Lead Eng.	Initial baseline specification for core ERP modules and basic auth.
v1.5	2026-10-15	Arch Team	Incorporated 9-Stage Kanban lifecycle and GST invoice calculation engine.
v2.0	2026-10-21	Core Lead	Complete IEEE-830 specification for dedicated EWMS branch: Section 4 Employee Self-Service (ESS), 5-Tier RBAC, and Zero-Trust IP Firewall.

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1. INTRODUCTION

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to provide an unambiguous, rigorous, and verifiable definition of the **Enterprenex EWMS (Enterprise Workflow Management System)**. This document defines the capabilities, system architecture, external interfaces, performance guarantees, and security constraints required for enterprise deployment. It acts as the single source of truth for software engineers, QA teams, project managers, and executive leadership.

1.2 Scope of the Product

Enterprenex EWMS is an institutional, multi-tenant capable, secure web platform designed to unify all operational and governance aspects of modern technology enterprises. Unlike fragmented point tools, EWMS centralizes:

- **Employee Self-Service (ESS):** Attendance, breaks, regularization, leave approvals, timesheets, and payslips.
- **Human Capital & Compliance:** Employee master, KYC verification, onboarding agreements, and performance OKRs.
- **Workforce Execution:** 9-stage Kanban task progression, sprint velocity tracking, and hours accounting.
- **Enterprise Finance:** GST-compliant tax invoicing, partial payment tracking, expense claims, and vendor accounting.
- **Zero-Trust Security:** IP-whitelisted access, mandatory first-time password resets, and immutable audit logging.

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
EWMS	Enterprise Workflow Management System; the core software platform.
ESS	Employee Self-Service; the portal through which staff manage attendance, leaves, and requests.
RBAC	Role-Based Access Control; security mechanism restricting resource access according to authorization.
MoM	Minutes of Meeting; formal record of proceedings, decisions, and assigned action deliverables.
KYC	Know Your Customer / Know Your Contributor; verification of PAN, Aadhaar, and identity documents.
SLA	Service Level Agreement; target turnaround timeframe for IT tickets and task reviews.
IP Firewall	Institutional security layer blocking administrative routes unless originating from whitelisted IPv4 addresses.

2. SYSTEM ARCHITECTURE & PERSPECTIVE

2.1 System Architecture Topology

EWMS employs a decoupled, client-server architecture engineered for micro-latency performance, strict type safety, and enterprise reliability. The system is partitioned into three distinct layers:

1. **Presentation Layer (Single Page Application):** Built with React 19, TypeScript, and Vite. Implements a responsive layout featuring an adjustable 230px–480px sidebar, local storage state persistence, and modular sub-drawers for instant previews.
2. **Application & Business Logic Layer:** Node.js and Express RESTful services exposing strongly typed endpoints protected by JWT authentication, role guards, and IP restriction middleware.
3. **Data Persistence Layer:** PostgreSQL relational database managed through Prisma ORM, enforcing foreign key integrity, schema constraints, and ACID transactional guarantees.

2.2 Technology Stack Specification

Layer	Technology / Framework	Specification & Role
Frontend Core	React 19, TypeScript 5.8	Strict typing, component modularity, zero-runtime errors.
Build & Bundling	Vite 8.1 / Rolldown	Sub-second hot-module replacement and optimized asset chunking.
Styling & Theme	Vanilla CSS + Custom Properties	High performance CSS variables (--adm-*) supporting instant dark/light themes.
Backend API	Node.js 22 LTS, Express 4	Non-blocking asynchronous REST services, middleware pipeline.
Database & ORM	PostgreSQL 16, Prisma 6.2	Type-safe database client, schema migrations, foreign keys.
Security / Crypto	bcryptjs, jsonwebtoken, crypto	Salted password hashing (cost factor 10), signed token verification.

3. 5-TIER ROLE-BASED ACCESS CONTROL (RBAC) MATRIX

EWMS enforces strict segregation of duties through a 5-tier authorization hierarchy. Access to modules, data tables, and administrative actions is governed by the following authoritative matrix:

Enterprise Role	Primary Scope of Operation	Permitted Modules	Strict Limitations
Director / CEO	Total corporate governance, financial oversight, strategic decisions.	All 14 Modules (Full Read/Write, Financial Approvals, Strategy).	Subject to institutional audit logs and regulatory filing policies.
Manager	Sprint management, project delivery, team tasks, performance sign-offs.	Projects, Tasks (9-Stage Kanban), Team, Calendar, Invoicing, Clients.	Access restricted to assigned project teams and departmental scopes.
HR	Personnel records, talent recruitment, onboarding, leave balances, compensation.	Employee Master, Recruitment ATS, Leave Ledger, KYC Vault, MoM.	No access to IT infrastructure firewall or raw client billing margin details.
Employee	Self-service attendance, task execution, timesheets, personal requests.	Employee Workspace (Section 4 ESS), My Tasks, Documents Vault, Helpdesk.	Strictly zero access to peer salaries, client financials, or manager approvals.
Super Admin / IT Admin	Infrastructure security, IP restriction, API keys, system diagnostics.	Security & RBAC Firewall, Audit Logs, Integration Settings, Asset Requisitions.	Privileged operations logged immutably in tamper-proof security ledger.

4. FUNCTIONAL REQUIREMENTS: 14 ENTERPRISE MODULES

4.1 Module 1: Authentication & Password Governance

FR-AUTH-01 (Credential Validation): The system shall authenticate users via valid corporate email address or 12-digit Employee ID alongside bcrypt-hashed passwords.

FR-AUTH-02 (Mandatory First-Time Setup): Newly onboarded employees with flag mustChangePassword: true shall be forcibly redirected to /create-new-password and prohibited from accessing any operational module until a compliant password is set.

FR-AUTH-03 (Password Policy Enforcement): New passwords must satisfy minimum 8 characters, at least 1 uppercase letter, 1 numeric digit, and 1 special symbol.

FR-AUTH-04 (IP Whitelist Guard): Administrative and management routes (/admin/*) shall enforce IP firewall verification against approved enterprise CIDR blocks.

FR-AUTH-05 (Session Invalidation): Users logging out shall immediately invalidate their client-side token and session state.

4.2 Module 2: Employee Self-Service (ESS) & Workspace (User Manual Section 4)

In compliance with Section 4 of the official EWMS User Manual, the Employee Self-Service Suite provides five dedicated operational tabs:

FR-ESS-01 (Attendance Check-In / Check-Out - Section 4.1): The system shall allow employees to punch check-in and check-out with automatic timestamp recording, shift allocation (09:30 AM – 06:30 PM), and visual duty status indicator.

FR-ESS-02 (Break Recording - Section 4.1): The system shall provide a dedicated break tracker supporting 15-minute Tea Break and 45-minute Lunch Break with live running stopwatches and shift resumption.

FR-ESS-03 (Attendance Regularization - Section 4.1): In case of biometric scanner or network failure, employees shall submit a formal Regularization Request specifying date, punch type, and accurate explanation for manager review.

FR-ESS-04 (Leave Management & Balances - Section 4.2): The system shall display live balances for Paid Annual Leave (12 days), Casual Leave (6 days), and Sick Leave (5 days), allowing employees to file leave requests with half-day toggles and manager approval tracking.

FR-ESS-05 (Timesheets & Hours Logging - Section 4.3): Employees shall record daily hours against assigned client projects, categorizing hours as Billable or Internal, and submit weekly timesheets for manager audit sign-off.

FR-ESS-06 (Self-Service Documents & Requests - Section 4.4): The system shall provide instant self-service generation for:

- **Monthly Salary Slip (Payslip):** Formatted with corporate letterhead, PAN, PF, PT, TDS, Basic, HRA, and printable PDF download.
- **Expense Reimbursement:** Claims for travel, meals, SaaS tools, and hardware with project allocation.
- **Official HR Letters:** Requisition for Bonafide, Experience Certificate, Address Proof, and Visa NOC.
- **IT Hardware Allocation:** Requisitions for development laptops, 4K displays, and security tokens.

FR-ESS-07 (Goals & OKRs - Section 4.5): The system shall render individual Q4 OKR percentage progress bars and HR-validated technical competency ratings.

4.3 Module 3: Executive & Operations Dashboard

FR-DASH-01 (Role-Adaptive Views): Dashboard shall dynamically render the Director Strategic View, Manager Operational View, or Employee Workspace based on the active user clearance.

FR-DASH-02 (Financial Telemetry): For Director/Manager roles, calculate real-time Gross Revenue, Paid Receivables, Outstanding Debt, and GST liabilities from live invoice tables.

FR-DASH-03 (Sprint Velocity): Render monthly sprint velocity bar charts with percentage efficiency deltas.

4.4 Module 4: Client Accounts & CRM Pipeline

FR-CRM-01 (Client Master Directory): Maintain verified client records including company name, industry, GSTIN, primary contact, billing address, and SLA tier.

FR-CRM-02 (Sales Opportunity Stages): Track leads across 5 stages: Lead Identified → Contacted → Proposal Sent → Negotiation → Won/Closed.

4.5 Module 5: Project Portfolio & Milestone Tracking

FR-PROJ-01 (Project Lifecycle): Support project states: Planning → In Progress → In Review → Completed → On Hold.

FR-PROJ-02 (Budget & Cost Governance): Compare allocated monetary budgets against actual expense claims and billable employee timesheet hours.

FR-PROJ-03 (Milestone Delivery): Enable multi-stage milestone tracking with target completion dates and deliverable attachments.

4.6 Module 6: 9-Stage Agile Kanban Work Delivery Engine

FR-TASK-01 (Formal 9-Stage Lifecycle): In accordance with Section 7.3 of the EWMS User Manual, all tasks shall transition through the standardized 9-stage state machine:

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Backlog → To Do → In Progress → Blocked → In Review → Revision Required → Approved → Completed → Cancelled
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FR-TASK-02 (Audit Trail): Every status change, assignee update, and priority adjustment must record the author and timestamp.

FR-TASK-03 (Interactive Task Drawer): Clicking any task opens a preview drawer displaying description, checklist items, subtasks, work logs, and team discussion threads.

4.7 Module 7: Human Resources & Employee Master

FR-HR-01 (12-Digit Employee ID Generator): Auto-generate unique sequential 12-digit institutional employee IDs (e.g. 202600000004).

FR-HR-02 (KYC Document Vault): Securely store verified PAN numbers, Aadhaar numbers, and signed employment contracts with verification stamps.

FR-HR-03 (Public Verification API): Provide public credential verification at /verify for authorized background checkers without exposing internal company data.

4.8 Module 8: Finance, GST Invoicing & Payment Reconciliation

FR-FIN-01 (GST Tax Invoices): Generate official GST invoices with itemized descriptions, quantities, rates, 18% GST (CGST 9% + SGST 9% or IGST 18%), and subtotal calculations.

FR-FIN-02 (Partial Payments & Reconciliation): Support recording partial payments against invoices, retaining the outstanding balance and payment reference numbers.

FR-FIN-03 (Invoice Status Automation): Automatically mark invoices as Paid when the balance reaches zero, or Overdue when payment exceeds due dates.

4.9 Additional Enterprise Modules Summary (Modules 9 to 14)

FR-IT-01 (IT Assets): Hardware lifecycle management tracking asset ID, serial number, warranty expiry, and assigned employee.

FR-MEET-01 (Meetings & MoM): Calendar scheduling with automated Google Meet video links, WhatsApp broadcast notifications, and formal Minutes of Meeting (MoM) record storage.

FR-DOC-01 (Compliance Vault): Cryptographically secure storage for legal NDAs, client master agreements, and corporate policies.

FR-BI-01 (Reporting & Telemetry): Export operational registers to CSV, Microsoft Excel, and printable PDF documents.

FR-SEC-01 (Zero-Trust IP Whitelisting): Strict firewall rejecting non-whitelisted IP addresses attempting administrative access.

FR-SEC-02 (Audit Ledger): Immutable logging of every security-sensitive transaction (logins, role adjustments, password modifications, financial approvals).

5. NON-FUNCTIONAL REQUIREMENTS (NFRs)

Category	Requirement Specification	Target Metric / SLA
Performance	Initial page load time on standard broadband connections.	< 1.20 Seconds (Vite production bundle)
API Latency	Server response time for authenticated database queries.	< 200 ms (95th percentile)
Availability	System operational uptime excluding scheduled maintenance.	99.95% Availability
Data Security	Cryptographic protection of passwords and tokens at rest and in transit.	TLS 1.3 in transit • bcrypt (cost 10) at rest
IP Restriction	Firewall enforcement on administrative routes.	100% block rate on unauthorized CIDR IPs
Data Integrity	Relational integrity and audit-trail preservation.	Zero data loss; ACID transactional guarantees
Compliance	Statutory labor, tax, and personal data protection compliance.	DPDP Act 2023 • Indian PF/ESI/TDS Standards

6. VERIFICATION & SIGN-OFF

This Software Requirements Specification has been formally reviewed, validated against active production code in the EWMS repository branch, and approved by the undersigned authorities for enterprise rollout.

Role / Authority	Name & Designation	Date	Status
Lead Architect	Core Engineering Team	2026-10-21	**VERIFIED & CODE-ALIGNED**
Lead Project Manager	Jyothi Anbhule (Lead PM & Mentor)	2026-10-21	**APPROVED FOR SPRINT**
Managing Director / CEO	Rohit Pawar (Director / CEO)	2026-10-21	**FORMALLY RATIFIED**